

WASP-43b

R-2548

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140330 · created 2026-08-26T00:40:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456746.6381 +/- 0.00073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.058
Transit depth (Rp/Rs)^2	3.21 +/- 2.06 [%]
Orbital Inclination (inc)	78.56 +/- 2.87
Airmass coefficient 1 (a1)	0.43 +/- 0.23
Airmass coefficient 2 (a2)	0.98 +/- 1.12
Scatter in the residuals of the lightcurve fit is	57.09 %
Best Comparison Star	#1 - [115, 22]
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.023 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.179 $\pm$ 0.058 vs 0.1594 (deviation 0.26 σ)
Duration fit vs published	0.023 d vs 0.0512 d (deviation 1.67 σ)
Mid-time O-C	-1.9 min
Depth SNR	2.4
Residual scatter	57.09 %

Light curve

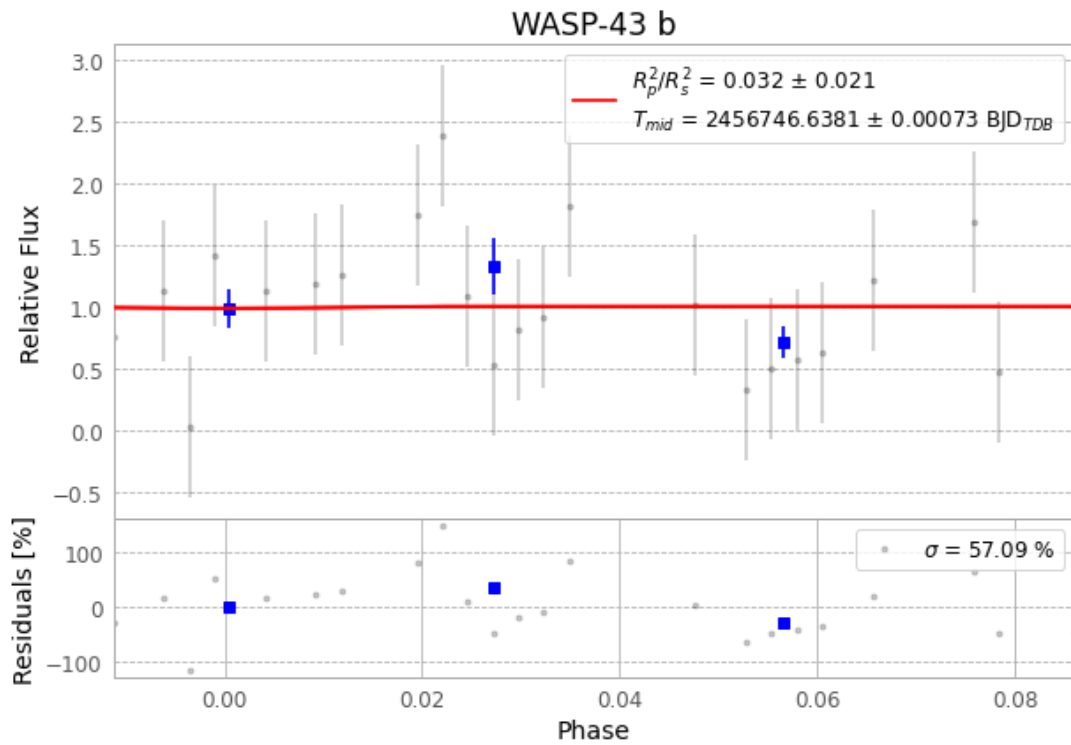


Figure 1 — FinalLightCurve_WASP-43 b_2014-03-29.png (SHA-256 69866013bd26dd66...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 de0a754fd9af726b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

42 FITS frames (28 MB), WASP-43140330024815.FITS → WASP-43140330045712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140330.log (572b563061fd...), FinalLightCurve_WASP-43 b_2014-03-29.png (69866013bd26...), FinalLightCurve_WASP-43 b_2014-03-29.csv (4ccf12ee6ad3...)

Compute resources

Wall time 87.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 00:40Z.